

State-Dependent Macroeconomic Forecasting of S&P 500 Returns

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1 Motivation

The forecasting of the S&P 500 equity premium remains a fundamental challenge in empirical finance due to the extremely low signal-to-noise ratio (SNR) inherent in monthly stock returns. Traditional forecasting models often rely on ordinary least squares (OLS) or static linear frameworks, which suffer from two primary challenges:

- **Regime Instability:** Standard models assume constant coefficients, ignoring the propensity of macroeconomic variables to behave differently during financial stress regimes or recessions.
- **Revision Bias:** Conventional backtesting utilizes “final revised” data. In reality, macroeconomic indicators are revised months or years after their first release. Using revised data introduces a significant look-ahead bias that overstates predictive power.

This project addresses these issues by constructing a statistically rigorous, leak-free framework using **ALFRED (Archival FRED)** real-time vintages and a **State-Dependent Supervised Screening & Regularized Factor (SSRF)** architecture to identify conditional shifts in market behavior.

2 Methods

The project utilizes a multi-stage SSRF pipeline designed to extract target-guided signals while maintaining strict causality.

2.1 Data Acquisition & Integrity

We utilize the `fredapi` library to programmatically access ALFRED vintages.

- **Indicator Universe:** 128 indicators from the FRED-MD database, categorized into five economic buckets: (1) Output and Income, (2) Labor Market, (3) Inflation and Prices, (4) Interest Rates and Spreads, and (5) Consumption and Sentiment.
- **Point-in-Time Discipline:** For a forecast made at time t , only the data vintage available at $t - 1$ is utilized, ensuring that future revisions do not contaminate the training set.

2.2 The SSRF Architecture

To mitigate “noise reification” in low-SNR environments, we adopt a four-stage defensive pipeline re-computed solely on the training window for every walk-forward fold:

1. **Stage 1: Group-Wise Supervised Screening:** Predictors are screened within economic buckets to prevent category dominance. We retain predictors with univariate absolute t-statistics above a tunable threshold θ . If a bucket yields no features, it contributes zero factors.
2. **Stage 2: Predictive Scaling:** Retained predictors are scaled by their univariate predictive regression slopes to prioritize signal over variance.
3. **Stage 3: Supervised Factor Extraction:** PCA extracts a conservative number of latent factors ($K = 10$) from the screened/scaled set.
4. **Stage 4: Interaction Modeling:** Factor sensitivities are modulated by a **Causal Soft-Regime Proxy** ($P(Z_t)$), defined as the rolling 12-month volatility percentile computed relative to the training distribution. Factors are centered to ensure interaction orthogonality:

$$\tilde{F}_t = F_t - \bar{F}_{train}$$

$$r_{t+1} = \alpha + \beta \tilde{F}_t + \delta(\tilde{F}_t \cdot P(Z_t)) + \epsilon_t$$

where β represents the base macro effect and δ captures the conditional regime shift.

3 Intended Experiments

The model will be evaluated using a rigorous expanding walk-forward strategy.

- **Validation Architecture:** An initial training window of 120 months is used. Hyperparameter optimization (α , L_1 ratio, θ) is performed via **Nested TimeSeriesSplit** within the training block to preserve temporal order.
- **Statistical Benchmarking:** Performance will be measured via the Campbell-Thompson R^2_{OOS} relative to the historical mean. We apply the **Campbell-Thompson restriction**, truncating negative excess return forecasts to zero ($\hat{r}_{t+1} = \max(0, \hat{r}_{t+1})$).
- **Economic Utility:** We will assess the model on its ability to improve the **Calmar Ratio** and reduce **Maximum Drawdown** compared to a Buy-and-Hold benchmark.

4 Team

- **Jagat Brahma:** Lead on the ALFRED real-time data acquisition pipeline and FRED API integration. Developed the SSRF factor extraction engine, including the group-wise screening and predictive scaling modules. Integrated the ElasticNetCV interaction model and designed the expanding walk-forward validation and evaluation metrics.

References

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